

SIMATS ENGINEERING ASSESSMENT TOOL

COURSE NAME: Theory of Computation **COURSE CODE:** CSA13

COURSE OUTCOME COVERED

CO1: Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems.(BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks:25 Duration : 1 Hour

Analytical Problem

Code of Conduct:

I _Pusalapati Chandu_(192372119)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

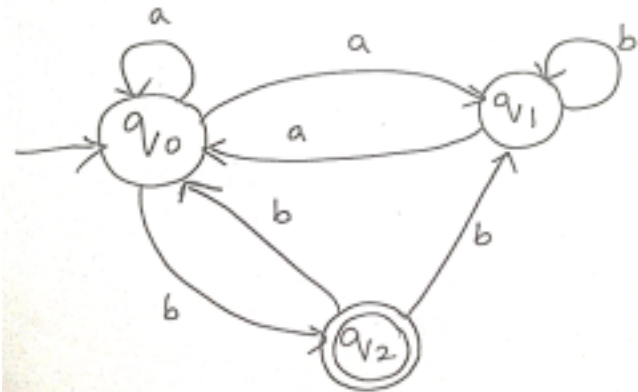
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Pchandu

SIMATS ENGINEERING ASSESSMENT TOOL

1. Consider the NFA given below which accepts some language L over the alphabet $\Sigma = \{a, b\}$. Construct a DFA that represents the same language L

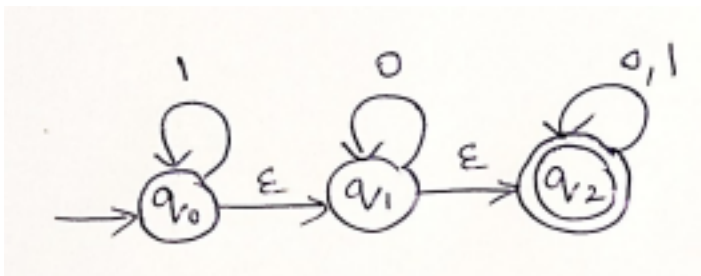


2. Let L be the language of words of length at least 2 whose last two letters are different from each other. For example $abaab$ is in L but not bb , Construct an NFA for L , Then draw the equivalent DFA

3. Design a Nondeterministic Finite Automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string **01011** is accepted by the NFA

4. Consider the NFA given below which accepts the language representing the set of all strings having 001 as a substring over the alphabet $\Sigma = \{0, 1\}$. Construct a DFA that represents the same language.

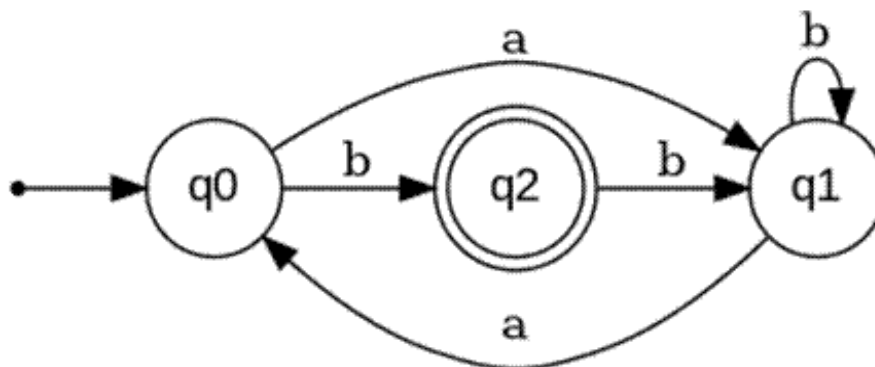
5. A NFA with ϵ -transitions is given below. Find ϵ -closure for each state and Design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final



states

Question 1 — NFA to DFA Conversion

Given NFA over $\Sigma = \{a, b\}$, states $\{q_0, q_1, q_2\}$, start state q_0 , final state q_2 :



Step 1: Formal transition table of the NFA

State	a	b
$\rightarrow q_0$	q_1	q_2
q_1	q_0	q_1
$* q_2$	—	q_1

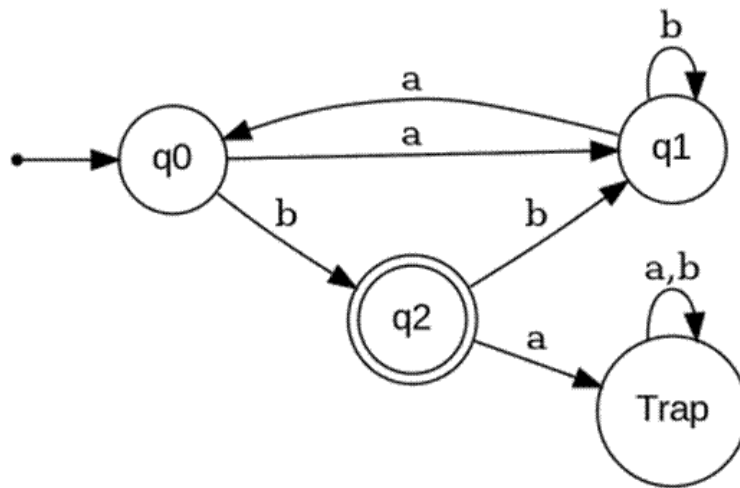
($\rightarrow$ denotes start state, $*$ denotes final state)

Step 2: Observation

Every state already has at most one transition per input symbol, so the automaton is deterministic except that q_2 has no transition defined on 'a'. This missing transition is what makes it technically an NFA (a partial function). To build a complete DFA we simply add one extra 'trap' (dead) state to absorb the missing transition.

Step 3: Equivalent DFA

State	a	b
$\rightarrow q_0$	q_1	q_2
q_1	q_0	q_1
$* q_2$	Trap	q_1
Trap	Trap	Trap



Final DFA = $(\{q_0, q_1, q_2, \text{Trap}\}, \{a, b\}, \delta, q_0, \{q_2\})$ — this accepts exactly the same language L as the original NFA.

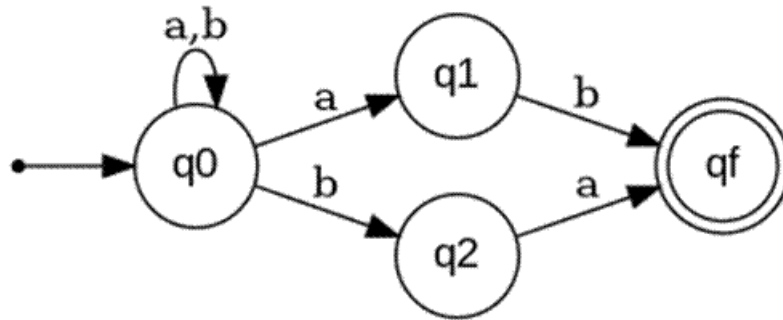
Question 2 — Last Two Letters Are Different

$L = \{ w \in \{a, b\}^* : |w| \geq 2 \text{ and the last two symbols of } w \text{ are different} \}$ (i.e. w ends in 'ab' or 'ba').

Step 1: NFA construction

Idea: stay in q_0 reading any prefix; nondeterministically 'guess' that the current symbol is the second-last symbol of the string, and check the following symbol differs from it.

State	a	b
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
q_1	—	$\{q_f\}$
q_2	$\{q_f\}$	—
$* q_f$	—	—

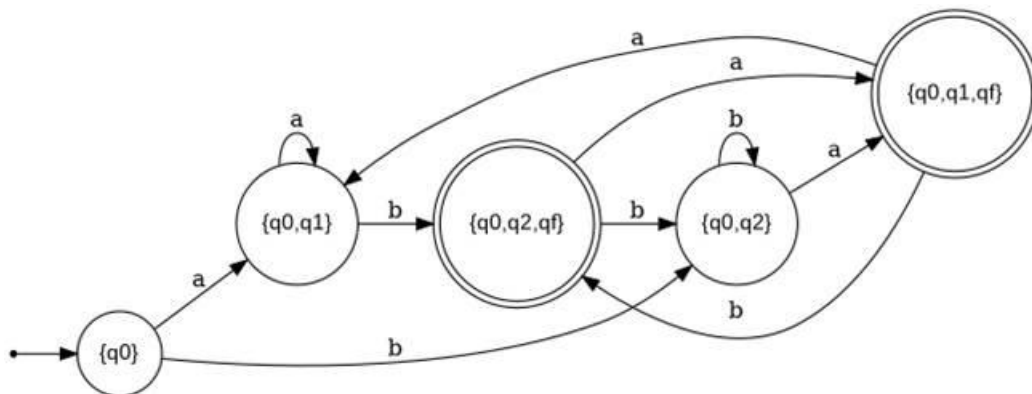


Step 2: Subset construction (NFA → DFA)

Let $S=\{q_0\}$, $A=\{q_0,q_1\}$, $B=\{q_0,q_2\}$, $C=\{q_0,q_2,qf\}$, $D=\{q_0,q_1,qf\}$.

DFA State	a	b
$\rightarrow S = \{q_0\}$	A	B
$A = \{q_0,q_1\}$	A	C
$B = \{q_0,q_2\}$	D	B
* $C = \{q_0,q_2,qf\}$	D	B
* $D = \{q_0,q_1,qf\}$	A	C

Final states are C and D (they contain qf). The resulting DFA has 5 states and is already reachable/minimal for this construction.



Verification

Check 'abaab' (should be accepted, ends in 'ab'): $S \xrightarrow{a} A \xrightarrow{b} C \xrightarrow{a} D \xrightarrow{a} A \xrightarrow{b} C$. Ends in C (final) ✓ accepted.

Check 'bb' (should be rejected): $S \xrightarrow{b} B \xrightarrow{b} B$. Ends in B (not final) ✓ correctly rejected.

Question 3 — Last Two Digits Are the Same (00 or 11)

Formal definition of the NFA

$N = (Q, \Sigma, \delta, q_0, F)$ where:

$Q = \{q_0, q_1, q_2, q_f\}$

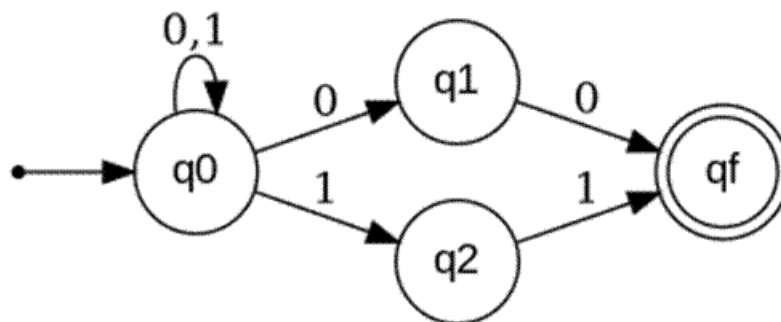
$\Sigma = \{0, 1\}$

start state = q_0

$F = \{q_f\}$

Transition function δ :

State	0	1
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
q_1	$\{q_f\}$	—
q_2	—	$\{q_f\}$
* q_f	—	—



Idea: q_0 loops on any input while nondeterministically guessing that the current symbol starts the last two digits; q_1 confirms the pair '00', q_2 confirms the pair '11'.

Showing the string 01011 is accepted

The string ends in '11', so we choose the path that guesses the pair at positions 4–5:

$q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_0 \xrightarrow{0} q_0 \xrightarrow{1} q_2 \xrightarrow{1} q_f$

Symbol by symbol:

Step	Symbol read	State reached
1	0	q0
2	1	q0
3	0	q0
4	1	q2 (guess: last pair begins here)
5	1	qf (accepting)

Since the automaton ends in $qf \in F$ after reading all 5 symbols, the string 01011 is accepted by the NFA. ✓

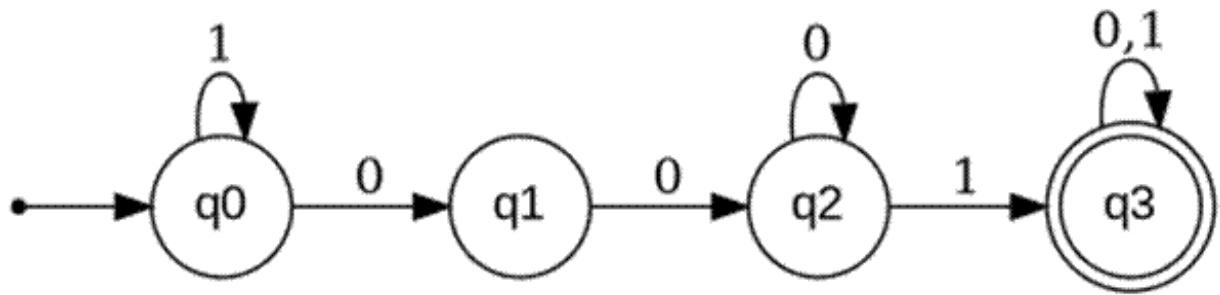
Question 4 — Strings Containing '001' as a Substring

Note: no diagram was supplied for this question in the assessment sheet, so the standard NFA for 'contains 001 as a substring' is constructed below and then converted to its DFA.

Step 1: NFA construction

Idea: q0 loops on any symbol until the machine nondeterministically guesses it has found the start of '001'; q1 confirms the first '0', q2 confirms '00', and q3 confirms '001' and stays accepting forever after.

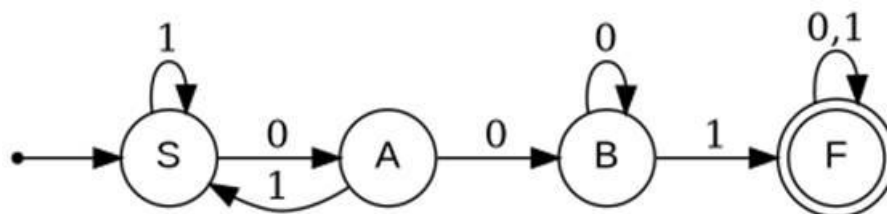
State	0	1
→ q0	{q0, q1}	{q0}
q1	{q2}	—
q2	{q2}	{q3}
* q3	{q3}	{q3}



Step 2: Subset construction

Let $S=\{q_0\}$, $A=\{q_0,q_1\}$, $B=\{q_0,q_1,q_2\}$, and once q_3 is reached the set always stays a superset containing q_3 . All such 'contains q_3 ' sets behave identically (once you've seen 001 you can never un-see it), so after merging equivalent accepting sets we get a single accepting state F :

DFA State	Meaning	0	1
$\rightarrow S$	no progress yet	A	S
A	seen '0'	B	S
B	seen '00'	B	F
* F	seen '001' (absorbing)	F	F



This 4-state DFA is the minimal DFA equivalent to the given NFA and accepts exactly the strings over $\{0,1\}$ that contain '001' as a substring.

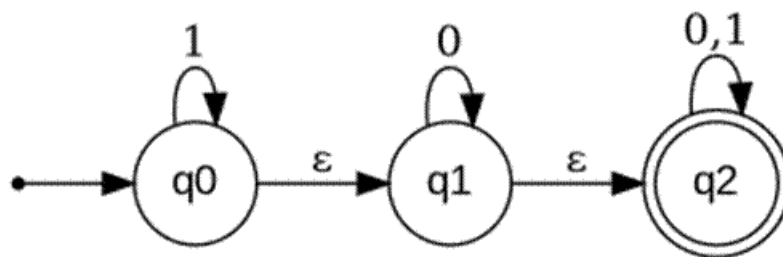
Verification

Check '1100101' (contains 001 ✓): S-1->S-1->S-0->A-0->B-1->F-0->F-1->F. Ends in F ✓ accepted.

Check '1010' (no 001): S-1->S-0->A-1->S-0->A. Ends in A (not accepting) ✓ correctly rejected.

Question 5 — ϵ -NFA to DFA (ϵ -closure method)

Given ϵ -NFA over $\Sigma = \{0,1\}$, states $\{q_0, q_1, q_2\}$, start q_0 , final q_2 :



State	0	1	ϵ
$\rightarrow q_0$	—	$\{q_0\}$	$\{q_1\}$
q_1	$\{q_1\}$	—	$\{q_2\}$
$* q_2$	$\{q_2\}$	$\{q_2\}$	—

Step 1: ϵ -closure of each state

State	ϵ -closure
q_0	$\{q_0, q_1, q_2\}$
q_1	$\{q_1, q_2\}$
q_2	$\{q_2\}$

Step 2: Subset construction using ϵ -closures

Start state of DFA = ϵ -closure(q_0) = $\{q_0, q_1, q_2\} \rightarrow$ call it A. Since A contains q_2 , A is a final state (accepts the empty string too).

A = $\{q_0, q_1, q_2\}$: on 0 $\rightarrow$ move(A,0) = $\{q_1, q_2\}$, ϵ -closure = $\{q_1, q_2\}$ = B. on 1 $\rightarrow$ move(A,1) = $\{q_0, q_2\}$, ϵ -closure = $\{q_0, q_1, q_2\}$ = A.

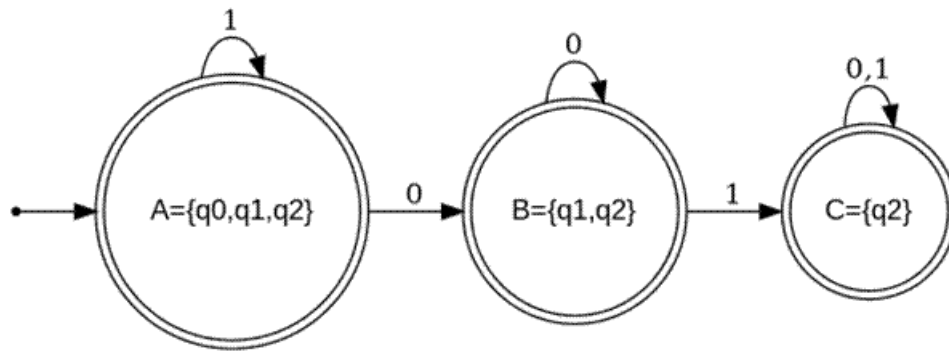
B = $\{q_1, q_2\}$: on 0 $\rightarrow$ move(B,0) = $\{q_1, q_2\}$, ϵ -closure = $\{q_1, q_2\}$ = B. on 1 $\rightarrow$

$\text{move}(B, 1) = \{q_2\}$, $\varepsilon\text{-closure} = \{q_2\} = C$.

$C = \{q_2\}$: on 0 $\rightarrow \{q_2\} = C$. on 1 $\rightarrow \{q_2\} = C$.

Step 3: DFA transition table

DFA State	NFA states	0	1	Final?
$\rightarrow A$	$\{q_0, q_1, q_2\}$	B	A	Yes
B	$\{q_1, q_2\}$	B	C	Yes
C	$\{q_2\}$	C	C	Yes



Conclusion

Every reachable DFA state contains q_2 , so every state is a final state. This means the DFA (and hence the original ε -NFA) accepts every string over $\{0, 1\}$, including the empty string — i.e. $L = \Sigma^*$.